
# *TRAINING LAB. 6*

# *TOPOLOGY OPTIMIZATION OF MOTORCYCLE FRAME*

The frame of the motorcycle shown in Fig.1 is a space frame made by welded beams. At first approximation, the frame can be modelled by a truss structure.

*Fig. 1 Motorcycle frame structure*

The frame supports the main loads applied to the motorbike during operation.

The most severe conditions are acceleration and braking.

The frame performances can be evaluated in terms of total mass and structural stiffness.

The aim of this laboratory is to determine the optimal structural configuration that minimizes the compliance for the structure given a constraint on the maximum volume. The compliance of the structure has to be minimized considering all the possible loading conditions.

The truss structure of Fig.2 is considered, where 17 nodes are defined and the elements given by all the possible connections among the nodes are introduced. Boundary conditions are imposed on nodes 9 and 17. The structure of Fig.2 is often denoted as “ground structure” and will be used as a starting solution for the topology optimization problem.

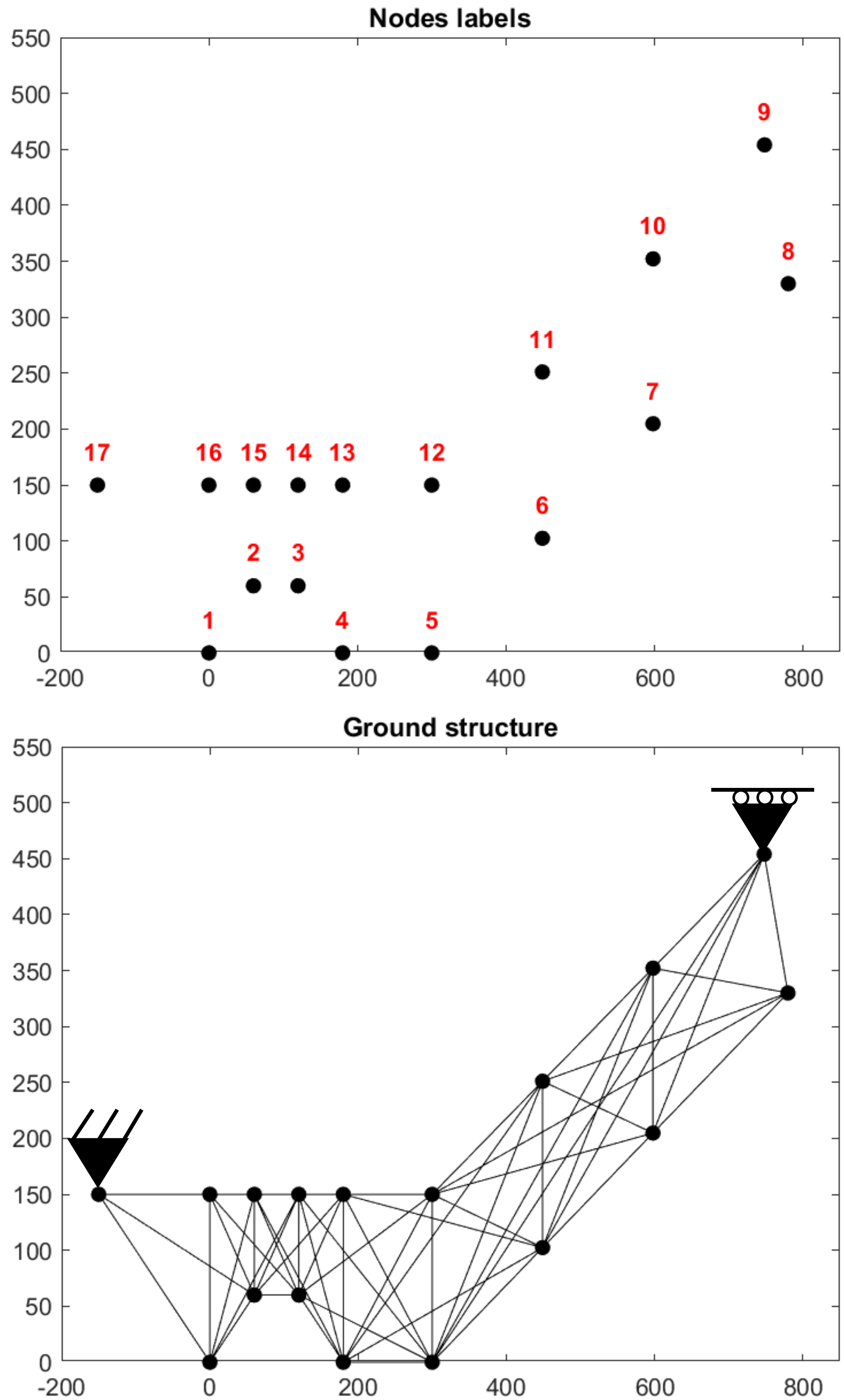


*Fig. 2 Truss structure representing the nodes and the ground structure of the frame*

The frame is made of steel with the following characteristics:

- Elastic modulus E = 210'000 N/mm$^2$
- Density $\rho$ = 7.7·10$^{-6}$ kg/mm$^3$
- Cross sectional area lower bound, lb = 0.001 mm$^2$
- Cross sectional area upper bound, ub = 700 mm$^2$

Nodal coordinates and element connectivity table are given in the form of text files uploaded on WeBeep, along with the Matlab function that solves the Finite element problem.
The two loading conditions (acceleration and braking) are given in the following table.

| **Loadcase 1 (braking)** | | **Loadcase 2 (acceleration)** | |
|---|---|---|---|
| **Node 8** | Fx = -13000 N<br>Fy = +2000 N | **Node 1** | Fx = +2000 N<br>Fy = -8000 N |
| **Node 9** | Fx = +9000 N<br>Fy = 0 N | | |

The provided MATLAB code computes the nodal displacements and stresses in the structural members by applying the finite element approach.

Since we are dealing with multiple loadcases, the objective function (to be minimized) can be the weighted sum of the compliances of each load case:

$$C = w_1 \mathbf{F}_1^T \mathbf{u}_1 + w_2 \mathbf{F}_2^T \mathbf{u}_2$$

Where $w_1$ and $w_2$ are two weights such that $0 \le w_1 \le 1$ and $w_2 = 1 - w_1$.

With:

$$\mathrm{lb} \le x_i \le \mathrm{ub}$$

$$V \le V_0 = \alpha\, V_{max}$$

Where $V_{max}$ is the maximum volume of the ground structure, i.e. the volume of the structure of Fig. 2 when all $x_i$ = ub, while $\alpha$ is the volume fraction.

**Requests**

- Find the optimal structure for $\alpha$ = 0.1 and $w_1 = w_2 = 0.5$. Plot the convergence diagram (compliance against iterations) and show the displacements of the loaded nodes for the obtained solution;
- For the volume fraction defined above, analyze and comment the results obtained by varying the weights in the range [0, 1].

## Tutorial

The computational scheme is the same as the previous lab. The only differences are:

- The objective function is a weighted sum of the compliances for the two considered load cases:

$$\mathrm{C} = \mathrm{w}_1\, \mathrm{C}_1 + \mathrm{w}_2\, \mathrm{C}_2 = \mathrm{w}_1\, \mathbf{F}_1^{\mathrm{T}} \mathbf{u}_1 + \mathrm{w}_2\, \mathbf{F}_2^{\mathrm{T}} \mathbf{u}_2$$

- The resulting sensitivity is a weighted sum of the two sensitivies:

$$\frac{\partial \mathrm{C}}{\partial \mathrm{x}_\mathrm{i}} = \mathrm{w}_1 \frac{\partial \mathrm{C}_1}{\partial \mathrm{x}_\mathrm{i}} + \mathrm{w}_2 \frac{\partial \mathrm{C}_2}{\partial \mathrm{x}_\mathrm{i}}$$

  With:

$$\frac{\partial \mathrm{C}_\mathrm{j}}{\partial \mathrm{x}_\mathrm{i}} = -\mathbf{u}_{\mathrm{i,j}}^{\mathrm{T}}\, \mathbf{k}_\mathrm{i}^0\, \mathbf{u}_{\mathrm{i,j}}$$